

Functions of Several Variables & Partial Derivatives

ANSWER KEY



Evaluating functions of two variables

- 1 For $f(x, y) = x^2y - 3xy + y^2$, find $f(2, -1)$.
 $f(2, -1) = (4)(-1) - 3(2)(-1) + 1 = -4 + 6 + 1 = 3$.
- 2 For $f(x, y) = \sqrt{x^2 + y^2}$, find $f(3, 4)$ and describe the domain of f .
 $f(3, 4) = \sqrt{9 + 16} = 5$. Domain: all of $\mathbb{R}^2$, since $x^2 + y^2 \geq 0$ always.

Computing partial derivatives

- 3 For $f(x, y) = x^4y^3 + 2xy - y^5$, find f_x and f_y .
 $f_x = 4x^3y^3 + 2y$. $f_y = 3x^4y^2 + 2x - 5y^4$.
- 4 For $f(x, y) = e^{x^2y}$, find f_x and f_y .
 $f_x = 2xy e^{x^2y}$. $f_y = x^2 e^{x^2y}$.

Second-order and mixed partial derivatives

- 5 For $f(x, y) = x^3y^2$, find f_{xx} , f_{yy} , and f_{xy} .
 $f_x = 3x^2y^2$, $f_y = 2x^3y$. $f_{xx} = 6xy^2$. $f_{yy} = 2x^3$. $f_{xy} = 6x^2y$.

Clairaut's Theorem: equality of mixed partials

- 6 For $f(x, y) = \sin(xy) + x^2y$, verify that $f_{xy} = f_{yx}$.
 $f_x = y \cos(xy) + 2xy$, so $f_{xy} = \cos(xy) - xy \sin(xy) + 2x$. $f_y = x \cos(xy) + x^2$, so $f_{yx} = \cos(xy) - xy \sin(xy) + 2x$. Equal, as expected.
- 7 State Clairaut's Theorem and the condition it requires.
If f_{xy} and f_{yx} are both continuous on an open region, then $f_{xy} = f_{yx}$ throughout that region.

Domains and level curves

- 8 Find the domain of $f(x, y) = \ln(x - y)$.
Requires $x - y > 0$, i.e. $x > y$. Domain: $\{(x, y) : x > y\}$.
- 9 Describe the level curve $f(x, y) = 4$ for $f(x, y) = x^2 + y^2$.
 $x^2 + y^2 = 4$ is a circle of radius 2 centered at the origin.

More partial derivative practice

- 10 For $f(x, y) = x^3 - 4xy^2 + y^3$, find f_x and f_y .

$$f_x = 3x^2 - 4y^2. \quad f_y = -8xy + 3y^2.$$

- 11 For $f(x, y) = \ln(x^2 + y^2)$, find f_x and f_y .

$$f_x = \frac{2x}{x^2 + y^2}. \quad f_y = \frac{2y}{x^2 + y^2}.$$

Functions of three variables

- 12 For $f(x, y, z) = x^2yz^3$, find f_x , f_y , and f_z .

$$f_x = 2xyz^3. \quad f_y = x^2z^3. \quad f_z = 3x^2yz^2.$$

- 13 For $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, find $f(1, 2, 2)$.

$$f(1, 2, 2) = \sqrt{1 + 4 + 4} = \sqrt{9} = 3.$$

More second-order partials practice

- 14 For $f(x, y) = e^{xy}$, find f_{xx} , f_{yy} , and f_{xy} .

$$f_x = ye^{xy}, \quad f_y = xe^{xy}, \quad f_{xx} = y^2e^{xy}, \quad f_{yy} = x^2e^{xy}, \quad f_{xy} = e^{xy} + xye^{xy} = (1 + xy)e^{xy}.$$

Level curves as a family of curves

- 15 Describe the family of level curves $f(x, y) = c$ for $f(x, y) = y - x^2$, as c varies.

$y - x^2 = c \Rightarrow y = x^2 + c$ — a family of upward parabolas, each identical to $y = x^2$ but shifted vertically by c .

Domains involving multiple restrictions

- 16 Find the domain of $f(x, y) = \frac{\sqrt{9 - x^2 - y^2}}{y}$.

Needs $9 - x^2 - y^2 \geq 0$ (inside or on the circle of radius 3) AND $y \neq 0$ (denominator). Domain:
 $\{(x, y) : x^2 + y^2 \leq 9, y \neq 0\}$.

One more round of partial derivative practice

- 17 For $f(x, y) = \frac{x}{y}$, find f_x and f_y .

$$f_x = \frac{1}{y}, \quad f_y = -\frac{x}{y^2}.$$

- 18 For $f(x, y) = \cos(x + 2y)$, find f_x and f_y .

$$f_x = -\sin(x + 2y), \quad f_y = -2\sin(x + 2y).$$

A closing pair: evaluation and partials together

- 19** For $f(x, y) = x^2 e^y$, find $f(2, 0)$, $f_x(2, 0)$, and $f_y(2, 0)$.
 $f(2, 0) = 4e^0 = 4$. $f_x = 2xe^y$, so $f_x(2, 0) = 4$. $f_y = x^2 e^y$, so $f_y(2, 0) = 4$.
- 20** Describe the level surface $f(x, y, z) = 1$ for $f(x, y, z) = x^2 + y^2 + z^2$.
 $x^2 + y^2 + z^2 = 1$ is a sphere of radius 1 centered at the origin.
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A final pair on mixed partials

- 21** For $f(x, y) = x^2 \sin y$, find f_{xy} and f_{yx} , and confirm they match.
 $f_x = 2x \sin y$, so $f_{xy} = 2x \cos y$. $f_y = x^2 \cos y$, so $f_{yx} = 2x \cos y$. They match, as Clairaut's Theorem predicts.
- 22** For $f(x, y) = \frac{y}{x}$ ($x \neq 0$), find f_{xx} .
 $f_x = -\frac{y}{x^2}$. $f_{xx} = \frac{2y}{x^3}$.
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Sketching a family of level curves

- 23** Sketch (describe) the level curves $f(x, y) = c$ for $f(x, y) = y - x^3$, for $c = -1, 0, 1$.
 $y - x^3 = c \Rightarrow y = x^3 + c$ — each is the cubic $y = x^3$ shifted vertically by c : for $c = -1$, $y = x^3 - 1$; for $c = 0$, $y = x^3$; for $c = 1$, $y = x^3 + 1$. All three are congruent cubic curves, stacked vertically.
- 24** For the same $f(x, y) = y - x^3$, find f_x and f_y , and verify ∇f is never 0.
 $f_x = -3x^2$, $f_y = 1$. Since $f_y = 1$ always, $\nabla f = \langle -3x^2, 1 \rangle$ can never be $\langle 0, 0 \rangle$ — consistent with the level curves never having a corner or self-intersection.
- 25** For $f(x, y, z) = xyz + x^2$, find f_x , f_y , f_z , and the second-order partial f_{xz} .
 $f_x = yz + 2x$. $f_y = xz$. $f_z = xy$. $f_{xz} = \frac{\partial}{\partial z}(yz + 2x) = y$.
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Piecing together a full evaluation problem

- 26** For $f(x, y) = x^2 + 2xy - y^2 + 3x - 4$, find $f(1, -2)$, $f_x(1, -2)$, and $f_y(1, -2)$.
 $f(1, -2) = 1 - 4 - 4 + 3 - 4 = -8$. $f_x = 2x + 2y + 3$, so $f_x(1, -2) = 2 - 4 + 3 = 1$. $f_y = 2x - 2y$, so $f_y(1, -2) = 2 + 4 = 6$.
- 27** For the same $f(x, y) = x^2 + 2xy - y^2 + 3x - 4$, find f_{xx} , f_{yy} , and f_{xy} .
 $f_{xx} = 2$. $f_{yy} = -2$. $f_{xy} = 2$.
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A boundary-behavior question

- 28** Find the domain of $f(x, y) = \sqrt{x} + \sqrt{y}$, and describe what happens to f as (x, y) approaches a boundary point like $(0, 4)$.

Domain: $\{(x, y) : x \geq 0, y \geq 0\}$ (the first quadrant, including its axes). At $(0, 4)$, $f(0, 4) = 0 + 2 = 2$ — the function is still defined and continuous there, since the domain includes its boundary.